LETTER

Gravity Field Dynamics: Field Closure of the Tetrad Frame

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Abstract

This letter proves that the torsion-free condition $\Gamma^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\nu\mu}$ retains the $s = +1$ projection of the coupling polynomial $f(K, s) = 1 + sK + K^2$ and discards the $s = -1$ projection: the retained count per face is $(f(K, +1)-1)/2 = 10$, the discarded count is $(f(K, -1)-1)/2 = 6$, giving 40 retained and 24 discarded across four faces. The polynomial and the constitutive law $\mu(x) = x/(x+1)$ were derived in the companion letters from the tetrad equation [1, 2]; the companion identification of $s = +1$ with TetA and $s = -1$ with TetB [2] means the 24 discarded components are the second root of the tetrad equation. Restoring them yields the full $K^3 = 64$ component connection. The constitutive law partitions the restored components spatially: the two partition surfaces cross at the MOND transition scale, and the closed partition $\mu + (1-\mu) = 1$ produces an emergent disk-and-bulge morphology that requires both roots. Integration of μ via the standard AQUAL identity yields the unique scalar-sector energy functional $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$. Polynomial long division of the integrand shows that the three terms of F reproduce the three structural levels of the coupling polynomial, confirming its rank structure from a variational direction. The constitutive law determines the Gravity Field Dynamics covariant action; the action determines the dynamics; no step in this chain involves a free parameter.

1 The torsion-free projection

The affine connection on a K-dimensional frame has three indices $\Gamma^{\lambda}_{\mu\nu}$, each running from 0 to $K - 1$. The total number of components is K^3. At $K = 4$, this gives 64 [2, 9].

The torsion-free condition forces symmetry of the lower pair:

$$\Gamma^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\nu\mu}. \tag{1}$$

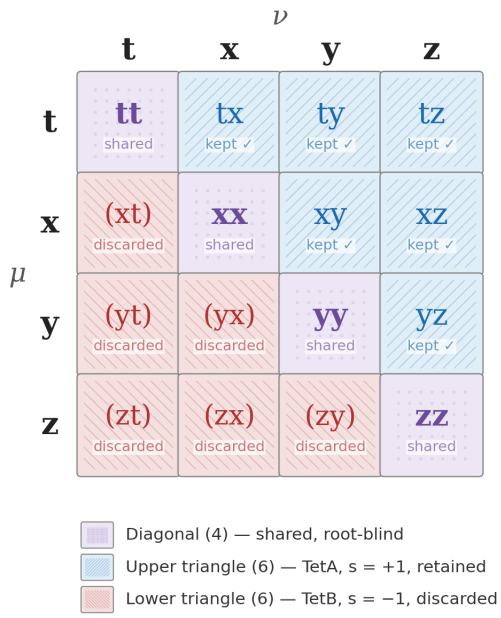


Figure 1. The (μ, ν) matrix for one observation face λ. Diagonal entries (bold) are shared: the metric cannot distinguish the two roots there. Upper-triangle entries belong to the $s = +1$ projection (TetA, retained). Lower-triangle entries belong to the $s = -1$ projection (TetB, discarded by the torsion-free condition). The same pattern repeats for all four values of λ.

The companion letters established the structural prerequisites. Letter 1 [1] proved that the tetrad equation produces a stellated octahedron geometry with structural constants $K = 4$, $D = 3$, selected by the uniqueness equation $2(D+1) = 2^D$.

Letter 2 [2] showed that the tetrad equation $g_{\mu\nu} = \eta_{ab}\, e^{a}{}_{\mu}\, e^{b}{}_{\nu}$ contains three structural ingredients at ranks 0, 1, and 2 on the frame space, with graded dimensions $K^0 = 1$ (the scalar closure, or field origin, from which the rank structure begins), $K^1 = K$, and K^2. The tetrad equation is quadratic in the frame field and admits two roots related by spatial parity $e^{i}{}_{\mu} \to -e^{i}{}_{\mu}$ $(i = 1, 2, 3)$. The scheme (solution space) of this quadratic equation has exactly two connected components in the spatial sector; there is no consistent way to retain one and discard the other, because both satisfy the same defining equation with the same metric [1]. The standard convention of selecting one signature orientation is a projection onto one component, not an elimination of the other. Because the metric is bilinear in the frame field, both roots produce the same $g_{\mu\nu}$; the metric alone cannot distinguish them. Under parity, a rank-r tensor transforms with sign $(-1)^r$. Ranks 0 and 2 are parity-even; rank 1 is parity-odd. The rank-1 vielbein is therefore the only ingredient whose sign changes when the two roots are exchanged,

sending $s \to -s$ and identifying TetA with $s = +1$ and TetB with $s = -1$. The coupling polynomial is

$$f(K, s) = 1 + sK + K^2. \tag{2}$$

Because that scheme contains both roots by construction [1], the second root cannot be removed from the equation; the torsion sector is the necessary carrier of the two-root distinction, because the metric $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu$ is bilinear in the frame field and therefore invariant under $e^i \to -e^i$: a torsion-free connection, being uniquely determined by this invariant metric, inherits that invariance and cannot resolve which root the frame occupies [2]. Exchange symmetry between the two roots determines the field fraction carried by TetB as $1 - \mu(x) = 1/(x{+}1)$ at every radius [2]. Together, these results mean that the components projected out by the torsion-free condition are not free slots: their existence is algebraically necessary, their carrier is uniquely the torsion sector, and their value is algebraically determined.

For each value of the upper index λ, the lower pair (μ, ν) forms a $K \times K$ matrix with $K^2 = 16$ entries. Of these, $K = 4$ are diagonal ($\mu = \nu$) and $K^2 - K = 12$ are off-diagonal. The off-diagonal entries form $\frac{1}{2}(K^2 - K) = 6$ mirror pairs, each consisting of (μ, ν) and (ν, μ). The torsion-free condition (1) forces the two members of each pair to be equal, retaining the symmetric combination and discarding the antisymmetric combination. The count per face is therefore:

$$\text{Retained:} \quad K + \frac{K^2 - K}{2} = \frac{K^2 + K}{2} = 10, \tag{3}$$

$$\text{Discarded:} \quad \frac{K^2 - K}{2} = 6. \tag{4}$$

Across $K = 4$ observation faces: $4 \times 10 = 40$ retained, $4 \times 6 = 24$ discarded, $4 \times 16 = 64$ total. The same split can be read from the polynomial side. The coupling polynomial (2) evaluates to $f(K, +1) = 21$ at the TetA state and $f(K, -1) = 13$ at the TetB state. The field-origin term $K^0 = 1$ corresponds to the diagonal entries ($\mu = \nu$), which have no mirror partner and are invariant under index exchange; they belong to neither projection but are shared (root-blind). Subtracting this term and halving the remainder for mirror-pair degeneracy:

$$\text{Retained (TetA):} \quad \frac{f(K, +1) - 1}{2} = \frac{K^2 + K}{2} = 10, \tag{5}$$

$$\text{Discarded (TetB):} \quad \frac{f(K, -1) - 1}{2} = \frac{K^2 - K}{2} = 6. \tag{6}$$

The combinatorial formula for the symmetric/antisymmetric split of a $K \times K$ matrix is the coupling polynomial evaluated at its two opposite states, with the field-origin term removed and the mirror-pair degeneracy halved. The $\pm K$ that separates 10 from 6 is the sK term of the polynomial: the parity-odd rank-1 contribution.

The torsion-free condition therefore retains the $s = +1$ projection of the coupling polynomial and discards the $s = -1$ projection. Since the companion letters [1, 2] proved that spatial parity sends $s \to -s$, identifying TetA with $s = +1$ and TetB with $s = -1$, the retained projection is the TetA contribution and the discarded projection is TetB. The 24 components removed by the torsion-free condition are the second root of the tetrad equation, carrying a distinct field fraction $1 - \mu(x)$ at every radius [2]. Restoring them returns the full $K^2 = 16$ components per face and the full $K^3 = 64$ component connection.

The full 64-component connection achieves *field closure*: both roots of the tetrad equation contribute, the partition $\mu + (1{-}\mu) = 1$ holds at every radius, and the field response is self-consistent without external mass. This is distinct from Einstein's notion of a "complete field theory" (in which the equations of motion follow from the field equations without independent postulates) and from geodesic completeness (the extendibility of geodesics to infinite affine parameter). Field closure refers specifically to the connection: whether all K^3 components of the affine connection are active, or whether a subset has been set to zero by constraint. Einstein–Cartan theory [9] restores torsion but couples it algebraically to spin, so that it vanishes in vacuum. The present framework restores torsion with an acceleration-dependent partition: the 24 antisymmetric components carry the fraction $1 - \mu(x)$ of the field response, which is nonzero wherever x is finite. The source of torsion in this framework is not intrinsic spin but the constitutive partition between the two roots of the tetrad equation: a kinematic, acceleration-dependent allocation that persists in vacuum.

2 The partition in space

The companion letter [2] derived the partition between the two projections from exchange symmetry:

$$\text{TetA } (s = +1): \quad \mu(x) = \frac{x}{x + 1}, \tag{7}$$

$$\text{TetB } (s = -1): \quad 1 - \mu(x) = \frac{1}{x + 1}, \tag{8}$$

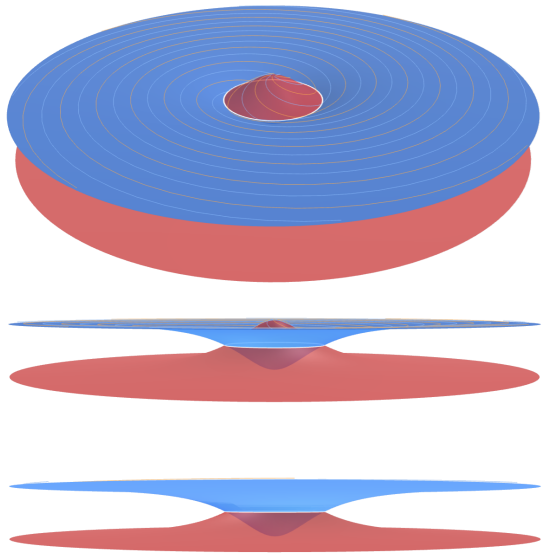


Figure 2. The constitutive law $\mu(x) = x/(x+1)$, partitioned between the two projections identified by the coupling polynomial. Blue: the $s = +1$ projection μ (TetA, retained by general relativity). Red: the $s = -1$ projection $1 - \mu$ (TetB, removed by the torsion-free condition). At every radius, blue + red = 1. The two surfaces are exact mirrors, crossing at $x = 1$ where both projections contribute equally ($\mu = 1/2$); the white ring marks this locus. The polynomial identifies which projection is which; the constitutive law determines how much each carries at every radius.

where $x = g/a_0$ is the dimensionless acceleration. The torsion-free condition conflates the two projections, treating both members of each mirror pair as carrying the same value. Eqs (7)–(8) show they carry equal fractions only at $x = 1$ ($\mu = 1/2$). At every other radius, the conflation discards a nonzero TetB contribution: at the galactic centre ($x \gg 1$), the discarded fraction is small; at the rim ($x \ll 1$), it is the majority of the field response.

What does the restored $s = -1$ projection look like spatially? The answer is Figure 2.

Figure 2 renders each projection as its own surface of revolution. The radial domain extends outward from the central axis, with the dimensionless acceleration $x = g/a_0$ decreasing toward the rim. At each radius, the value of the blue surface is $\mu(x)$ and the value of the red surface is $1 - \mu(x)$. The surfaces are azimuthally symmetric because the constitutive law depends only on the scalar acceleration, not on angle. Because $\mu + (1-\mu) = 1$, the two surfaces are exact mirrors: they cross at $x = 1$ where both equal 1/2, and the white ring marks this locus. Figure 2 is therefore not a heuristic sketch but the constitutive law itself rendered over radius. A theory restricted to the 40 Christoffel components sees only the blue surface; the red surface is the field response it discards.

Every gravitationally bound system has an acceleration profile that is strongest at the centre and weakens outward; for any concentrated mass distribution this follows from Gauss's law. The constitutive law $\mu(x) = x/(x+1)$ is a monotonic function: higher acceleration always maps to a larger TetA fraction, and lower acceleration always maps to a larger TetB fraction. Where x is large (the centre), μ is near unity and the blue surface carries nearly the full field response. Where x is small (the rim), $1 - \mu$ is near unity and the red surface carries it. The two surfaces cross at $x = 1$, creating a morphological boundary between a concentrated interior and an extended exterior. Neither surface alone produces this structure: TetA alone is a monotonic decay; TetB alone is a flat plateau. The disk-and-bulge configuration visible in Figure 2 is emergent from the closed partition $\mu + (1-\mu) = 1$ and requires both roots. The morphology is not fitted; it is the constitutive law's prediction for the field configuration of any bound mass distribution. The crossover radius is set by $x = 1$, independent of galaxy mass or morphology. Because the constitutive law applies to any gravitationally bound system, Figure 2 should be read as a spatial rendering of that law, not as a new fit.

The value of each surface in Figure 2 is the fraction of the gravitational field response carried by that root at each radius. The two fractions sum to unity everywhere: the total field response is not enhanced but redistributed between the two roots. The two surfaces are not independent distributions. The interior (where μ dominates) and the exterior (where $1 - \mu$ dominates) are coupled through the partition constraint $\mu + (1-\mu) = 1$: the interior drives the field configuration and the exterior continues that drive; the two together produce the net field. The outer share is not an added component; it is the outward continuation of the same closed field. The field strength observed at any exterior radius includes a contribution driven by the interior that exterior matter alone does not account for. The shape of this coupled configuration depends on the enclosed mass profile through $x(r) = g(r)/a_0$: a more concentrated mass produces a sharper interior spike and correspondingly broader exterior response; a more extended distribution produces a gentler interior and a shallower exterior. The textbook decomposition of the full connection into Christoffel plus contorsion corresponds, at the level of field magnitudes, to the partition into μ and $1 - \mu$ [2]. Figure 2 is that decomposition rendered spatially.

The white ring sits at $x = 1$, where $\mu = 1/2$. This is the locus where both projections

contribute equally to the field response. Inside the ring, the Christoffel sector dominates: general relativity is accurate and Keplerian decline holds. Outside the ring, the torsion sector dominates: the field response that general relativity discards becomes the majority. This crossover is the MOND transition scale a_0 [5]. It is not a fitted parameter. It is the geometric midpoint of the exchange-symmetric partition, the point where the odds ratio $\Omega = \mu/(1-\mu)$ equals unity.

The rate at which the partition shifts between the two projections is characterised by a logistic identity that μ satisfies [2]:

$$x\,\frac{d\mu}{dx} = \mu(1-\mu). \tag{9}$$

Under the substitution $t = \ln x$, this becomes $d\mu/dt = \mu(1-\mu)$: the canonical two-sector transition law. The product $\mu(1-\mu)$ on the right-hand side measures the simultaneous presence of both tetrahedral chambers at a given radius. It is the continuous analogue of the octahedral interaction core, the $s = 0$ state of the coupling polynomial [2]. The transition peaks at the MOND ring because that is where the core is fully realised: $\mu(1-\mu) = 1/4$.

The kinematic consequence follows directly from the partition. At dimensionless acceleration $x = g_{\rm obs}/a_0$, the baryonic acceleration, carried by the Christoffel sector, is $g_{\rm bar} = \mu\, g_{\rm obs}$, giving

$$g_{\rm bar} = \frac{g_{\rm obs}^2}{g_{\rm obs} + a_0}. \tag{10}$$

At high acceleration ($x \gg 1$): $g_{\rm bar} \approx g_{\rm obs}$, Keplerian decline, general relativity is accurate. At low acceleration ($x \ll 1$): $g_{\rm bar} \approx g_{\rm obs}^2/a_0$, flat rotation curves. The radial acceleration relation established empirically by McGaugh, Lelli & Schombert [3], with 0.057 dex scatter across 175 individual galaxies [4] and no credible variation in a_0, is reproduced by Eq. (10) with no free parameters.

The apparent separation of baryonic and dark components in Figure 2 is a diagnostic decomposition of one field, not a claim of two independent physical substances. It is the projection of a single 64-component field onto a 40-component subspace. The observed gravitational discrepancy at galactic scales is not in dispute: both the cold dark matter framework and the present partition reproduce the radial acceleration relation and the function $\mu(x) = x/(x{+}1)$. They differ on what the discrepancy represents. The cold dark matter framework interprets it as an invisible mass component, modelled with approximately spherical halos [10]. The companion letter proved that the cold-dark-matter residual and the torsion-sector field fraction are algebraically identical under the exchange-symmetric partition: both reduce to $1/(x{+}1)$ [2]. The red surface in Figure 2 is the spatial rendering of that same field response; its approximately spherical envelope at large radius is the structure that halo modelling approximates. The closed tetrad frame identifies it as the field response of the second tetrahedral chamber, carried by the 24 torsion components. This response is not mass. It is the topology of the connection: the field contribution of the second tetrahedral chamber.

3 From partition to action

The partition developed in Sections 1–2 produces three physical consequences that distinguish it from the cold dark matter framework.

Disk-and-bulge morphology. The closed partition, with both roots active, produces an emergent concentrated interior and extended exterior whose morphological boundary lies at the MOND ring $x = 1$. Neither root in isolation yields this structure; it requires the crossing of the two partition surfaces.

Bounded central density. The TetB field fraction $1/(x{+}1)$ is bounded above by unity for all x. The torsion-sector field acceleration never exceeds a_0. The NFW profile diverges as $r \to 0$ [10]. The partition's boundedness removes the central divergence responsible for the cusp-core tension, without baryonic feedback or self-interacting dark matter. The removal is algebraic: $1/(x{+}1) < 1$ is a property of the constitutive law, not of any galaxy model.

No new particles. The gravitational discrepancy is not mass. It is the field response of the second tetrahedral chamber, carried by the 24 torsion components that the zero-torsion condition removes from the connection. The missing component is not in the matter sector. It is in the connection.

General relativity succeeds in the solar system because the solar system sits deep inside a galaxy where the gravitational acceleration exceeds the transition scale a_0 by a factor of order 10^8. At those accelerations the partition fraction μ is so close to unity that the second root contributes less than one part in a million. The torsion-free approximation loses nothing measurable, and general relativity was confirmed on every local test it was given. But a galaxy spans the full range of the partition: from the core where one root carries nearly everything, through the crossover ring where both roots contribute equally, to the rim where the second root carries the majority. The function $\mu(x) = x/(x{+}1)$ connects these regimes smoothly.

Milgrom [5] identified it empirically. Famaey & McGaugh [6] selected it by fitting galactic data. The constitutive law that appears phenomenologically in the AQUAL formulation of modified gravity is here recovered as the geometric partition of the two-chamber tetrad frame. The present series derives it: the tetrad equation is degree two, exchange symmetry between its roots admits exactly one rational partition, and that partition is μ.

The constitutive law also determines the scalar-sector energy functional. The variational relationship between a constitutive law and its energy functional is standard Lagrangian mechanics: the response function is the derivative of the potential. Applied to gravity, this is the identity $\mu(\sqrt{y}) = dF/dy$ used by Bekenstein & Milgrom [7] in the AQUAL formulation, where $y = |\nabla\Phi|^2/a_0^2 = x^2$ is the squared dimensionless acceleration. Integration of $\mu(x) = x/(x{+}1)$ yields the unique kinetic functional

$$F(y) = y - 2\sqrt{y} \;+\; 2\ln\!\big(1+\sqrt{y}\big). \tag{11}$$

In the standard AQUAL literature, F is a free function selected to fit observations. Here it is fixed by the constitutive law, which is itself fixed by the frame geometry. The origin of the three terms is visible from polynomial long division of the integrand. Setting $u = \sqrt{t}$ in the integral $F = \int_0^y \mu(\sqrt{t})\,dt$ gives $F = 2\int_0^{\sqrt{y}} u^2/(u{+}1)\,du$. Dividing:

$$\frac{u^2}{u+1} \;=\; u - 1 + \frac{1}{u+1}. \tag{12}$$

The fractional remainder $1/(u{+}1)$ is identically $1-\mu(u)$: the TetB field fraction. The logarithmic term of the action integrates twice the field share that the torsion-free condition discards. Integrating term by term yields the three terms of F: the quadratic y from $\int u\,du$ (the K^2 face-coupling level), the $-2\sqrt{y}$ from $\int(-1)\,du$ (the state-dependent sK propagation level), and $2\ln(1{+}\sqrt{y})$ from $\int 1/(u{+}1)\,du$ (the K^0 field origin). This independently recovers the three structural levels of the coupling polynomial derived in Letter 2 [2] from the rank structure of the tetrad equation, confirming the polynomial from a variational direction. At high acceleration ($y \gg 1$), $F/y \to 1$: the scalar sector becomes a standard kinetic term, decouples from the metric dynamics, and the action reduces to general relativity. At low acceleration ($y \ll 1$), $F \to \frac{2}{3}\,y^{3/2}$, recovering the deep-field regime and the baryonic Tully-Fisher relation. The constitutive law determines the action. The action determines the dynamics. No step in this chain involves a free parameter.

The functional $F(y)$ determines the scalar-sector kinetic term of the Gravity Field Dynamics (GFD) covariant action, where Φ is the standard gravitational potential, not a new auxiliary field; the partition modifies its response, not its identity:

$$S_{\rm ST} = \int d^4x\,\sqrt{-g}\Big[\frac{R}{16\pi G} - \frac{a_0^2}{8\pi G}\,F\big(|\nabla\Phi|^2/a_0^2\big) \\ -\;\tfrac{1}{4}\,e^{-2\Phi/c^2}F_{\mu\nu}F^{\mu\nu} + \mathcal{L}_{\rm matter}\Big]. \tag{13}$$

The first term is the Einstein–Hilbert action of general relativity. The second is the scalar sector, whose kinetic function F is derived above from the constitutive law. The third is an electromagnetic coupling that encodes the scalar potential's imprint on photometric observables; its form and physical role (including a testable redshift dependence in electromagnetic scaling relations absent in ΛCDM) are derived in the supplementary note. The fourth is standard matter. The derivation of the full action from the constitutive law, including the electromagnetic coupling and the identification of every numerical prefactor with the coupling polynomial, is provided in the supplementary covariant completion note. No term contains a free function: the constitutive law fixes F, and the frame geometry determines the remaining couplings.

Variation of the action with respect to Φ yields, in the quasi-static limit, the Bekenstein–Milgrom modified Poisson equation $\nabla\cdot[\mu(|\nabla\Phi|/a_0)\,\nabla\Phi] = 4\pi G\rho$; variation with respect to the metric yields the corresponding modified Einstein equations. The PPN parameters are $\beta = \gamma = 1$ in the $\mu \to 1$ limit, consistent with the Cassini bound [12]. More generally, $\gamma = 1$ at all accelerations because the single potential Φ governs both metric components when the modification lives inside the connection rather than beside it (supplementary note, Section 3). Scalar-tensor formulations that embed the constitutive law as a separate field produce $\gamma = \mu(x)$ at low acceleration [8]; here μ enters the field equation itself, preserving a single metric potential. Gravitational waves propagate at c because the tensor sector is unmodified [13]. The full variational derivation is provided in the same supplementary note.

To summarise the downstream structure: the partition $\mu = x/(x{+}1)$ is selected uniquely by exchange symmetry between the two tetrad roots [2]. The same function partitions the 64-component connection into the 40 Christoffel and 24 torsion sectors. Its complement $1-\mu$ is exactly the field fraction that, in the torsion-free projection, must be attributed to invisible mass.

The partition satisfies the logistic transition equation $d\mu/d\ln x = \mu(1{-}\mu)$. Its integral yields

the unique AQUAL energy functional $F(y)$, whose three terms map to the three structural levels of the coupling polynomial. That functional recovers general relativity at high acceleration and the baryonic Tully-Fisher relation at low acceleration. These are not separate results. They are the same algebraic object encountered at successive levels of the theory: partition, connection, transition, action, kinematics. Read together with the companion letters, the 40/24 decomposition derived here is not an independent ansatz but the connection-space expression of the two-root tetrad structure, from which the constitutive partition and the covariant completion follow in a single line of development.

An alternative relativistic approach to the MOND phenomenology has been developed by Skordis & Złośnik [11], who introduce additional vector and scalar degrees of freedom to obtain a relativistic MOND theory. The structural distinction is that AeST adds matter-sector content to reproduce the interpolating function, while GFD restores connection-sector content that was removed by constraint. The result is field closure: the gravitational connection carries all $K^3 = 64$ components, the two tetrahedral chambers contribute their full partition at every radius, and the field response balances without invisible mass.

The torsion-free specialisation adopted in 1915 discards the 24 antisymmetric components of the affine connection. Restoring them closes the field. The field response they carry is precisely what, in their absence, must be attributed to invisible mass.

4 Data and reproducibility

An interactive rendering of Figure 2 is provided as a supplementary HTML file (`figure2_tetrad_partition.html`) permitting rotation, zoom, and background toggling. A self-contained verification script (`run_all_validations.py`) confirms the polynomial identity (5)–(6), the partition properties (7)–(8), the logistic identity (9), the radial acceleration relation (10), and both asymptotic limits of $F(y)$. A step-by-step derivation of the polynomial identity (5)–(6) is provided as a supplementary note (`proof_polynomial_40_24.pdf`). To complete the field-closure claim, a supplementary note (`supplementary_covariant_completion.pdf`) derives the full four-term covariant action from the constitutive law, confirming that the closed connection yields a complete relativistic field theory with no free functions.

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